

Project 1B: Custom API Integration

SAP S/4HANA ERP to FinSight Financial Analytics Platform

Resubmission upgraded with runnable prototype and validation evidence

Repository	https://github.com/sachin-git01/project2
Project Category	Enterprise Technology / API Integration / Financial Analytics
Primary Deliverable	End-to-end integration framework design and runnable simulation
Source System	SAP S/4HANA
Target System	FinSight financial analytics platform

1. Executive Summary

This project designs and demonstrates a custom API integration framework that connects a client's SAP S/4HANA ERP system with a financial analytics platform. The solution covers API specifications, data extraction, transformation logic, validation, error handling, reconciliation, monitoring, and stakeholder communication.

The resubmission includes a working Python prototype that simulates a General Ledger integration flow. It reads SAP-style source records, validates master data references, transforms valid records into a canonical FinSight payload, routes invalid records into a dead letter queue, and generates a reconciliation report.

2. Problem Understanding

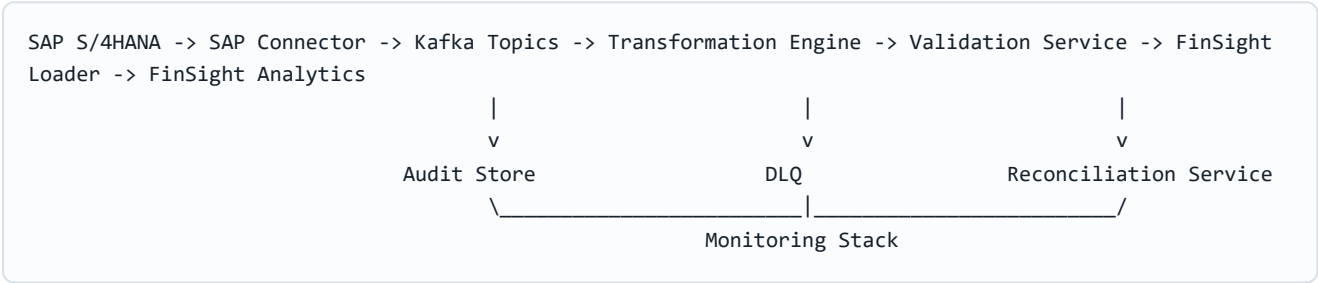
Manufacturing finance teams often rely on delayed exports from ERP systems, causing late visibility into journal postings, payables, receivables, cash, procurement, budget variance, and inventory valuation. The objective is to reduce manual data movement while preserving accounting accuracy, audit traceability, and SAP production stability.

3. Target Architecture

Layer	Design	Reason
SAP Extraction	ODP/CDS/OData/RFC/IDoc facade	Uses supported SAP integration patterns with delta tracking.

Messaging	Kafka topics per domain	Decouples SAP from FinSight and supports replay.
Transformation	Versioned rule engine	Maps SAP-specific fields into canonical financial schemas.
Validation	Mandatory field, date, currency, and master data checks	Prevents inaccurate financial analytics.
Loading	Idempotent FinSight batch upsert APIs	Prevents duplicate records during retry.
Reconciliation	Record count, totals, checksum, and lineage checks	Proves that every accepted record is accounted for.
Monitoring	Metrics, logs, traces, DLQ, freshness, and circuit breaker panels	Supports production operations and incident response.

Architecture Diagram



4. API Specification Coverage

The submission includes OpenAPI 3.0 specifications for SAP source extraction and FinSight destination ingestion.

API File	Coverage
api/API_SAP_Source.yaml	12 source endpoints covering GL, AP, AR, cost centre, profit centre, material ledger, purchase orders, sales orders, fixed assets, bank statements, budget/actual, and inventory.
api/API_FinSight_Destination.yaml	12 destination batch upsert endpoints plus ingestion status, OAuth 2.0 security, idempotency key header, rate limit headers, and shared error envelope.

5. Data Mapping and Transformation

The mapping specification contains 102 field-level mappings, exceeding the 50-field minimum. Mappings cover General Ledger, Accounts Payable, Accounts Receivable, Cost Centre, Profit Centre, Material Ledger, Purchase Orders, Sales Orders, Fixed Assets, Bank Statements, Budget vs Actual, and Inventory.

Rule Area	Implementation
Dates	SAP YYYYMMDD values are converted to ISO date format.
Currency	ISO currency validation and point-in-time conversion to INR.
Fiscal Periods	Periods 001-012 map normally; 013-016 are marked as special periods.
Idempotency	Target keys are generated from tenant, domain, company code, fiscal year, document number, and line item.
Lineage	Every target record keeps source system, source table, source document, and batch ID.

6. Error Handling and Resilience

Error Type	Handling
Transient SAP/API failures	Retry with exponential backoff and circuit breaker protection.
FinSight HTTP 429	Honor Retry-After header and throttle consumers.
Mapping failures	Route invalid records to DLQ with error code and source payload.
Duplicate writes	Use deterministic idempotency keys and payload hash comparison.
Reconciliation breaks	Generate break report and route to operations or finance data steward.

7. Runnable Prototype Evidence

The resubmission includes executable code under `src/sap_finsight_integration/`.

Prototype Component	File	Purpose
Extractor	<code>extract.py</code>	Reads SAP-style CSV records.
Transformer	<code>transform.py</code>	Validates and maps SAP rows to FinSight canonical records.
Loader	<code>load.py</code>	Writes simulated FinSight batch payload.
Reconciliation	<code>reconcile.py</code>	Calculates counts, debit/credit totals, net amount, checksum, and status.
CLI	<code>cli.py</code>	Runs the pipeline end to end.

Run Commands

```
$env:PYTHONPATH='src'  
python -m sap_finsight_integration.cli  
python -m unittest discover -s tests -p '*unittest.py'
```

Observed Validation Output

Batch: BATCH-GL-20260629-0930
Status: RECONCILED
Accepted: 6
Rejected/DLQ: 2
Ran 2 tests in 0.016s
OK

Reconciliation Report

```
{
  "batchId": "BATCH-GL-20260629-0930",
  "domain": "GENERAL_LEDGER",
  "sourceRecordCount": 8,
  "acceptedRecordCount": 6,
  "rejectedRecordCount": 2,
  "debitTotalInr": 253125.0,
  "creditTotalInr": 253125.0,
  "netAmountInr": 0.0,
  "checksum": "8676c1adfc4a3254d74b7435e4948929717a01668e919b648920480fca4665ef",
  "status": "RECONCILED"
}
```

Load Result

```
{
  "batchId": "BATCH-GL-20260629-0930",
  "status": "COMPLETED",
  "acceptedCount": 6,
  "rejectedCount": 0
}
```

8. Sample Input and DLQ Evidence

Sample SAP GL Input

```
company_code,fiscal_year,fiscal_period,document_number,line_item,gl_account,cost_centre,profit_centre,posting_date,currency,amount,debit_credit
1000,2026,003,5100001001,1,400000,CC100,PC100,20260628,INR,125000.00,S
1000,2026,003,5100001001,2,110000,CC100,PC100,20260628,INR,125000.00,H
1000,2026,003,5100001002,1,500000,CC200,PC200,20260628,USD,1000.00,S
1000,2026,003,5100001002,2,210000,CC200,PC200,20260628,USD,1000.00,H
1000,2026,013,5100001003,1,600000,CC300,PC300,20260628,EUR,500.00,S
1000,2026,013,5100001003,2,310000,CC300,PC300,20260628,EUR,500.00,H
1000,2026,003,5100001004,1,700000,CC999,PC100,20260628,INR,100.00,S
1000,2026,003,5100001005,1,700000,CC100,PC999,20260628,INR,100.00,H
```

DLQ Output

```
[
  {
```

```

    "batchId": "BATCH-GL-20260629-0930",
    "domain": "GENERAL_LEDGER",
    "sourceKey": "1000:5100001004:2026:001",
    "errors": [
      {
        "errorCode": "ERR-MAP-003",
        "field": "cost_centre",
        "message": "Unknown cost centre"
      }
    ],
    "sourcePayload": {
      "company_code": "1000",
      "fiscal_year": "2026",
      "fiscal_period": "003",
      "document_number": "5100001004",
      "line_item": "1",
      "gl_account": "700000",
      "cost_centre": "CC999",
      "profit_centre": "PC100",
      "posting_date": "20260628",
      "currency": "INR",
      "amount": "100.00",
      "debit_credit": "S"
    }
  },
  {
    "batchId": "BATCH-GL-20260629-0930",
    "domain": "GENERAL_LEDGER",
    "sourceKey": "1000:5100001005:2026:001",
    "errors": [
      {
        "errorCode": "ERR-MAP-003",
        "field": "profit_centre",
        "message": "Unknown profit centre"
      }
    ],
    "sourcePayload": {
      "company_code": "1000",
      "fiscal_year": "2026",
      "fiscal_period": "003",
      "document_number": "5100001005",
      "line_item": "1",
      "gl_account": "700000",
      "cost_centre": "CC100",
      "profit_centre": "PC999",
      "posting_date": "20260628",
      "currency": "INR",
      "amount": "100.00",
      "debit_credit": "H"
    }
  }
]

```

9. Testing Coverage

The full test plan includes 25 mandatory scenarios across functional, non-functional, failure, security, and reconciliation categories. The runnable prototype also includes automated unit tests for:

- Invalid master data routing to DLQ.
- Successful output generation.
- Balanced debit-credit reconciliation.
- Correct source, accepted, and rejected record counts.

10. Monitoring and Operations

Panel	Purpose
Pipeline Health	Shows green/yellow/red status per domain.
Throughput	Tracks records processed per second.
Error Rate	Alerts when errors exceed threshold.
DLQ Depth	Alerts when failed records require attention.
Reconciliation Status	Highlights financial breaks by domain.
SAP and FinSight Health	Tracks response time, pool utilization, rate limits, and tokens.

11. Deliverables Included

Deliverable	Artifact
Project overview	README.md
Requirements	docs/D0_Requirements_v1.md
Architecture	docs/D1_Integration_Architecture_v1.md and diagrams/*.mmd
API specification	docs/D2_API_Specification_v1.md and api/*.yaml
Mappings	mappings/MAP_Finance_AllDomains_v1.md
Error handling	docs/D4_Error_Handling_Framework_v1.md
Reconciliation	docs/D5_Reconciliation_Specification_v1.md
Monitoring	monitoring/D6_Monitoring_Dashboard_v1.md
Test plan	tests/D7_Integration_Test_Plan_v1.md
Runnable prototype	src/, data/sample/, outputs/, tests/test_pipeline_unittest.py
Single file bundle	submission_single_file.html

12. Conclusion

This resubmission addresses the project statement by combining a full enterprise integration design with runnable proof of the core pipeline behavior. The solution demonstrates API contract design, extraction planning, transformation rules, validation, DLQ handling, idempotency, reconciliation, monitoring, and stakeholder-ready documentation.